SIMATS ENGINEERING
ASSESSMENT TOOL 3

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C04-AT3

1. Fundamental Concepts Used in Web Development

XML (Extensible Markup Language)

XML is a markup language used to store, organize, and transport data in a structured format. It allows developers to create their own tags according to the type of information being stored.

Purpose and role:
- Stores data in a structured and readable format.
- Helps exchange data between different applications.
- Is platform-independent.
- Is commonly used for configuration files and data communication.

JSP (JavaServer Pages)

JSP is a server-side technology based on Java used to create dynamic web pages. JSP allows HTML pages to contain Java-based server-side functionality.

Purpose and role:
- Generates dynamic web content.
- Processes user requests on the server.
- Can communicate with databases.
- Commonly works with Servlets, JavaBeans, JSTL, and MVC architecture.

MVC Architecture

MVC stands for Model–View–Controller. It is a software architecture used to organize web applications into three components.
Model: Handles data, business logic, and database operations.
View: Displays information to the user, such as HTML or JSP pages.
Controller: Receives user requests and controls the flow between Model and View.
MVC separates application logic from presentation, making applications easier to maintain, test, and debug.

XSLT (Extensible Stylesheet Language Transformations)

XSLT is a language used to transform XML documents into other formats such as HTML, XML, or text.

Purpose and role:

• Converts XML data into a presentation format.
• Separates data from its presentation.
• Helps display XML data as web pages.
• Can select, sort, and transform XML elements.

XML Schema (XSD)

XML Schema, commonly called XSD (XML Schema Definition), defines the structure, elements, attributes, and data types allowed in an XML document.

Purpose and role:

• Validates XML documents.
• Defines the allowed structure of XML.
• Specifies data types and restrictions.
• Ensures data consistency and correctness.

JSTL (JavaServer Pages Standard Tag Library)

JSTL is a collection of standard tags used in JSP pages to perform common tasks without writing large amounts of Java code.

JSTL provides tags for conditions, loops, formatting, database-related operations, and XML processing. It makes JSP pages simpler, cleaner, and easier to maintain by reducing Java scriptlet code.

PHP (Hypertext Preprocessor)

PHP is a server-side scripting language used to develop dynamic and interactive web applications.

PHP can process form data, communicate with databases, manage sessions and cookies, generate dynamic HTML pages, and perform server-side calculations and business logic.

Database Connectivity

Database connectivity is the process of connecting a web application to a database so that it can store, retrieve, update, and delete data.

Common operations are Create/Insert, Read/Select, Update, and Delete. Java applications can use JDBC, while PHP applications can use database APIs such as PDO or MySQLi.

Database connectivity provides persistent storage and allows applications to manage user and application data.

Parser

A parser is a software component that reads and analyzes structured data according to a particular syntax or grammar. In web applications, XML parsers read XML documents and convert their contents into a form that programs can process.
DOM Parser loads the complete XML document into memory as a tree structure. SAX Parser reads XML sequentially and processes elements as they are encountered.
Parsers help applications read structured documents, validate syntax, extract data, and support communication between systems.

2. Interconnection of These Technologies in Building a Web Application

These technologies work together to create a complete web application. Each technology performs a specific function in processing, validating, storing, transforming, and presenting data.

Basic Flow

User → View → Controller → Model → Database → Model → View → User

Step 1: XML Structure

XML stores information using a hierarchical structure of elements and attributes. For example, student information can be stored using elements such as name, age, and department. XML provides a standard way to represent and exchange structured data.

Step 2: XML Validation Using XML Schema

An XML document can be validated against an XML Schema (XSD). The schema specifies which elements are allowed, their order, data types, and restrictions. Therefore, XML Schema ensures that the XML data follows the required structure before it is processed.

Step 3: Parsing XML

An XML parser reads the XML document and processes its contents. The parser allows the application to extract values such as a student's name, age, and department for further processing.

Step 4: Server-Side Processing Using JSP or PHP

JSP and PHP perform server-side processing. JSP uses Java-based server-side processing and commonly works with Servlets, databases, JSTL, and MVC. PHP processes requests, form submissions, business logic, and database operations on the server.

Step 5: MVC Architecture

MVC organizes the application into Model, View, and Controller. The Model communicates with the database and performs business operations. The View displays information to the user. The Controller receives requests, communicates with the Model, and selects the appropriate View.

Step 6: Database Connectivity

The Model communicates with the database through database connectivity mechanisms. The application can perform INSERT, SELECT, UPDATE, and DELETE operations. Retrieved data is passed back to the application for processing and display.

Step 7: XSLT for XML Transformation

When XML is used to represent application data, XSLT can transform XML into HTML or another format. Thus, XML + XSLT can produce a presentation that can be displayed in a web browser.

Step 8: JSTL in JSP

JSTL can be used within JSP pages to perform common operations such as looping through database records and applying conditions. This keeps presentation logic cleaner and reduces the need for Java scriptlet code.

Overall Interconnection

User Request → Controller (MVC) → Model / Business Logic → Database Connectivity → Database → Model → JSP / PHP View → JSTL or XSLT for presentation/transformation → Response to User

Conclusion

XML provides a structured method for storing and exchanging data, while XML Schema validates its structure. Parsers read and process XML documents, and XSLT transforms XML data into useful presentation formats. JSP and PHP perform server-side processing and generate dynamic web content. MVC organizes the application into Model, View, and Controller components, making it easier to maintain. JSTL simplifies JSP development, while database connectivity allows the application to store and retrieve persistent information. Together, these technologies form important building blocks for developing structured, dynamic, and maintainable web applications.

Code of Conduct:

I ______________________ (K.Deepika / 192371042) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Deepika

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand